

GenerativeTowns / SDFusion - Current Pipeline Report

Prepared by Codex on 2026-05-12.

The project now has a working OSM-to-3D pipeline and a separate generative research branch. The strongest current production path is OSM footprint/context -> retrieval or proposal image -> Hunyuan3D -> simplification -> height/placement correction -> quality audit. SDF remains in the pipeline as a geometric constraint, fit/audit representation, and future residual refinement layer.

Pipeline Overview

Current path:

OSM vector/map input -> footprint records -> retrieval/quality rerank or proposal image -> Hunyuan3D raw mesh -> simplify -> world placement -> area-aware heightfix -> quality audit.

SDF role:

- footprint correction and target signed-distance field
- fit scoring against target footprint
- geometry audit for mismatch and overhang
- possible residual SDF refinement branch
- possible abs(SDF)-as-UDF extraction for hollow/open shells

Current judgment: Hunyuan/retrieval carries visual mesh detail; SDF carries geometric constraint value.

Progress Snapshot

OSM corpus v2 summary:

```
```json
{
 "corpus_name": "campus_lafayette_v2",
 "dataset_version": "osm_generation_scaffold_v1",
 "count": 44,
 "positive_count": 37,
 "negative_count": 7,
 "classes": {
 "RESIDENTIALhouse": 41,
 "COMMERCIALoffice_building": 2,
 "PUBLICschool_building": 1
 },
 "splits": {
 "smoke12": 12,
 "north8": 8,
 "east6": 6,
 "west6": 6,
 "south6": 6,
 "northeast6": 6
 },
 "jsonl": "outputs/osm_generation_dataset_corpus_v2/campus_lafayette_v2_records.jsonl",
 "csv": "outputs/osm_generation_dataset_corpus_v2/campus_lafayette_v2_index.csv",
 "masks_npz": "outputs/osm_generation_dataset_corpus_v2/campus_lafayette_v2_footprint_masks.npz"
},
```
```

Proposal image dataset summary:

```
```json
{
 "dataset_version": "osm_proposal_image_v1",
 "name": "campus_lafayette_proposal_v1",
 "source_records": "outputs/osm_generation_dataset_corpus_v2/campus_lafayette_v2_records.jsonl",
 "count": 37,
 "train_count": 30,
 "val_count": 7,
 "classes": {
 "PUBLIC": 1,
 "RESIDENTIAL": 34,
 "COMMERCIAL": 2
 },
 "quality_statuses": {
 "pass": 37
 },
 "train_jsonl": "outputs/osm_proposal_image_dataset_v1/campus_lafayette_proposal_v1_train.jsonl",
 "val_jsonl": "outputs/osm_proposal_image_dataset_v1/campus_lafayette_proposal_v1_val.jsonl",
 "all_jsonl": "outputs/osm_proposal_image_dataset_v1/campus_lafayette_proposal_v1_all.jsonl"
},
```
```

Quality summary for the clean east2 run:

```
```json
{
 "count": 2,
 "pass": 1,
 "warn": 1,
 "fail": 0
},
```
```

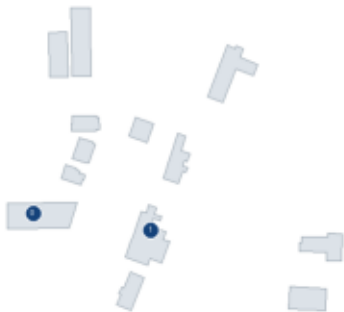
DDPM overfit summary:

```
```json
{
 "train_count": 8,
 "val_count": 8,
 "best_val_noise_mse": 0.022264983505010605,
 "ckpt_best": "outputs/osm_proposal_image_ddpm_v1_overfit8/ckpt_best.pth",
 "metrics_csv": "outputs/osm_proposal_image_ddpm_v1_overfit8/metrics.csv"
},
```
```

OSM Input to Placed Output

Map-only OSM input
outputs/osm_generative_proposal_clean_east2_heightfix/osm_map_input.png

OSM map input all footprints



Selected footprints
outputs/osm_generative_proposal_clean_east2_heightfix/osm_map_selected.png

selected footprints for pipeline



Map output with houses
outputs/osm_generative_proposal_clean_east2_heightfix/osm_map_output_houses.png

output map + placed houses



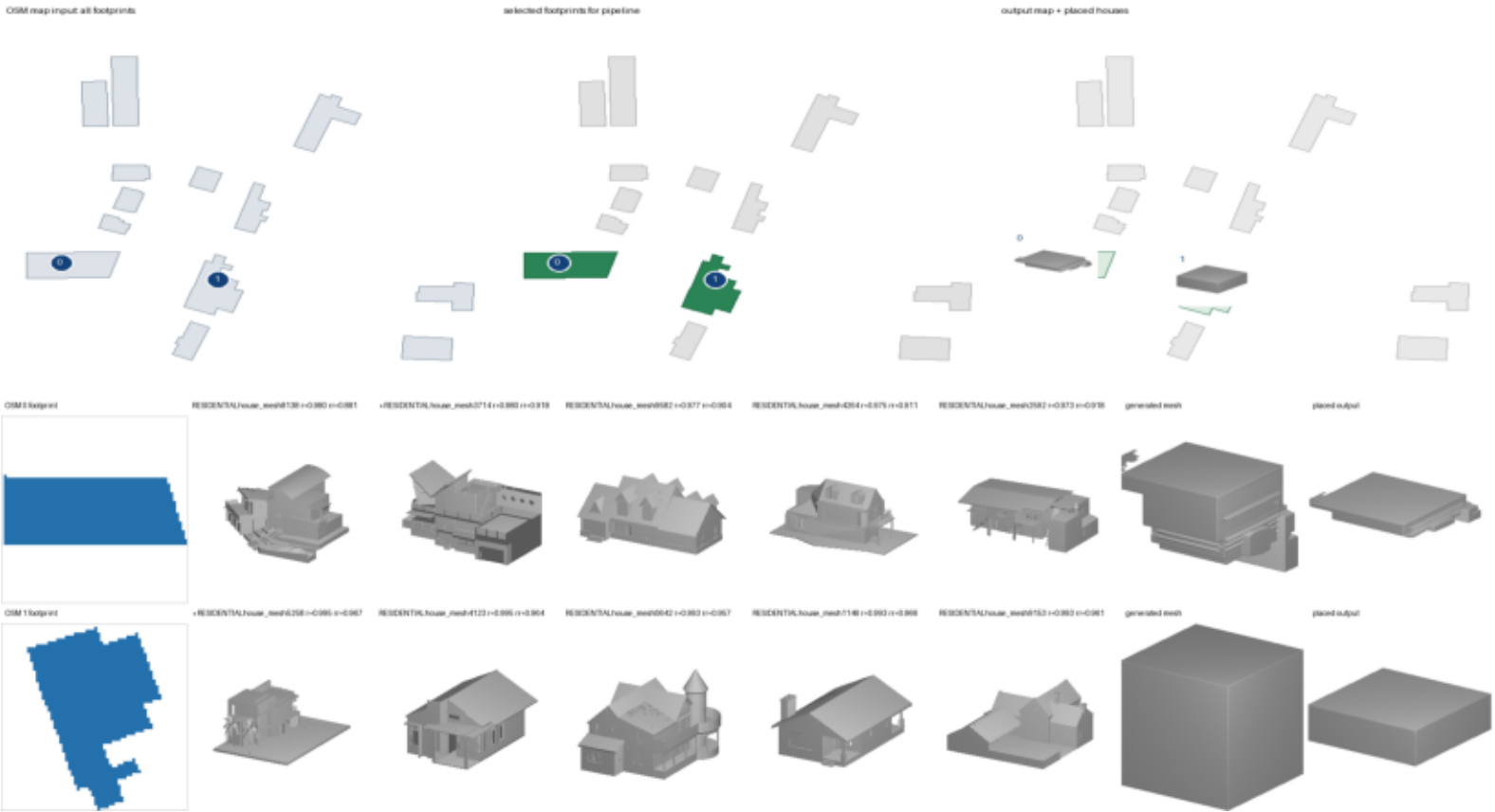
Rendered placed scene
outputs/osm_generative_proposal_clean_east2_heightfix/osm_hunyuan_scene_render.png

rendered output scene (3D)

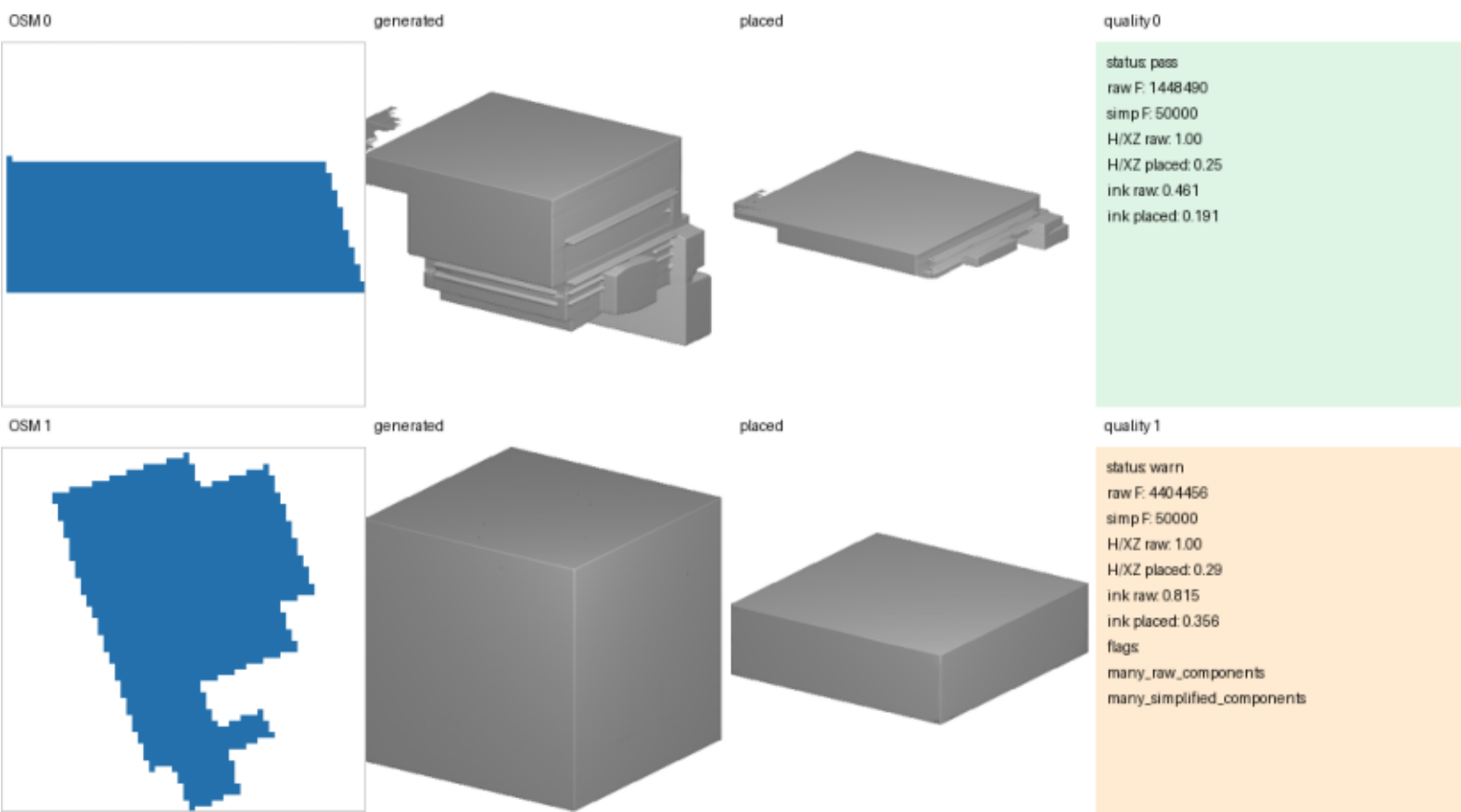


Candidate Choices and Quality Audit

Retrieval/proposal choices
outputs/osm_generative_proposal_clean_east2_heightfix/osm_map_choices_sheet.png

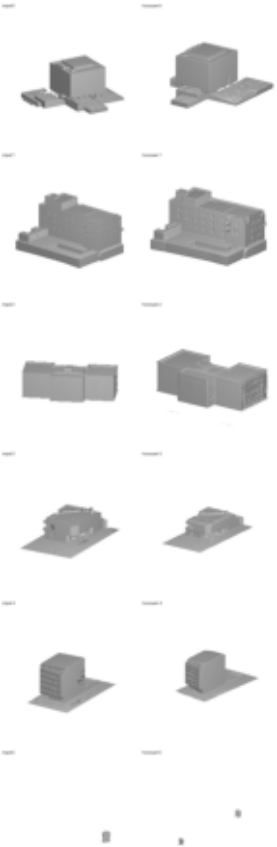


Generation quality audit
outputs/osm_generative_proposal_clean_east2_quality/generation_quality_audit_sheet.png

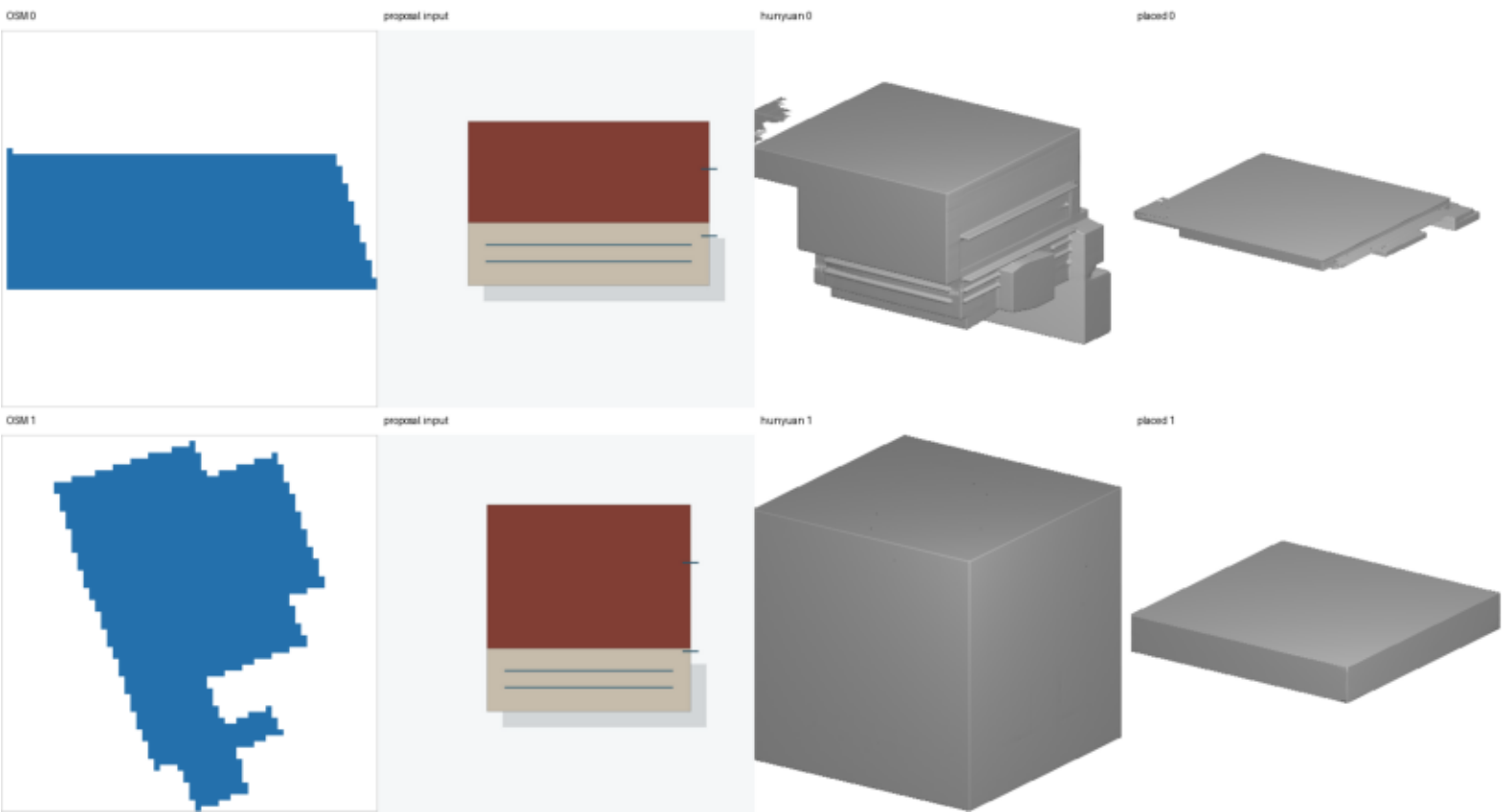


Hunyuan and Retrieval Evidence

Hunyuan retrieval rank-1 mini sheet
outputs/hunyuan_retrieval_rank1_mini/hunyuan_building_smoke_sheet.png

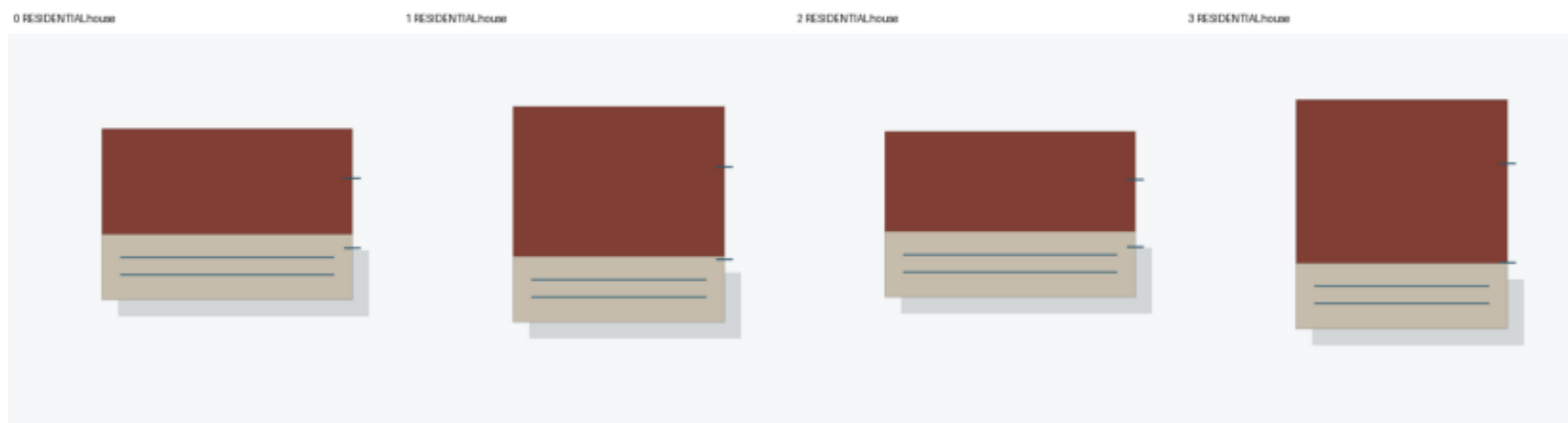


Clean east2 Hunyuan pipeline sheet
outputs/osm_generative_proposal_clean_east2/osm_hunyuan_pipeline_sheet.png



Proposal Image Models

Proposal input sheet
outputs/osm_generative_proposal_clean_east4/proposal_inputs_sheet.png



Supervised proposal preview, sharp loss
outputs/osm_proposal_image_generator_v1_sharp/val_preview_epoch_060.png

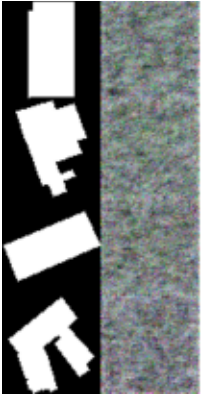


Supervised continued preview
outputs/osm_proposal_image_generator_v1_continued/val_preview_epoch_030.png

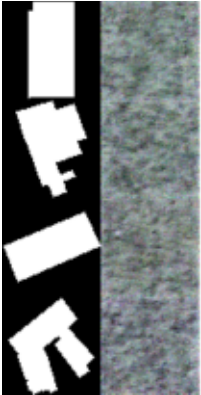


DDPM Diagnostic

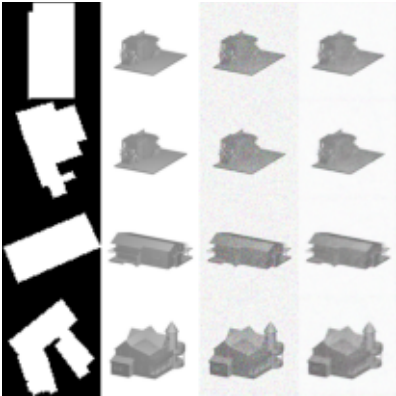
DDPM overfit epoch 300
outputs/osm_proposal_image_ddpm_v1_overfit8/sample_epoch_300.png



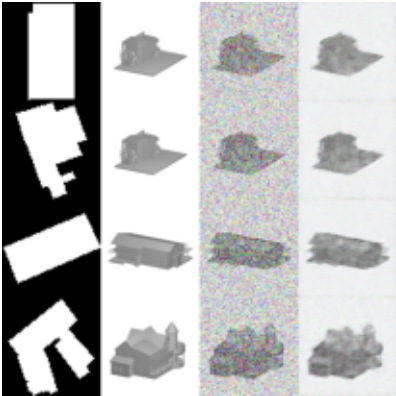
DDIM pure sample from resumed checkpoint
outputs/osm_proposal_image_ddpm_v1_overfit8_ddim_diag/sample_ddim_resume_epoch_300.png



DDIM reconstruction from t=25
outputs/osm_proposal_image_ddpm_v1_overfit8_ddim_diag/recon_t025_epoch_300.png



DDIM reconstruction from t=100
outputs/osm_proposal_image_ddpm_v1_overfit8_ddim_diag/recon_t100_epoch_300.png



Recommendation

Which paper direction suits our purpose?

ControlCity-style conditioning is the closest match, but implemented through a practical pretrained/latent diffusion backbone rather than a tiny scratch-trained pixel DDPM. The pipeline should use OSM footprint/context/height/class channels, keep SDF as a geometric constraint and audit layer, generate cleaner proposal images, and pass those images to Hunyuan/image-to-3D.

Immediate path: keep retrieval/supervised proposal generation for usable output quality.
Research path: controlled latent diffusion with SDF/footprint constraints.